

MIDTERM EXAM



Name: _____

ID: _____

Answer the following

Model (B)

1. (2 points) To implement an "Undo" feature in a word processor, which data structure would you use? *Stack*

2. (2 points) What is wrong in the following code?

```
void main(){  
    Stack s;    int i;  
    Push (&s, 5);    Pop(&s, &i);  
    printf("%d", i);  
}
```

The stack is not initialized
(No create function)

3. (2 points) Suppose we have a circular array implementation of the queue, with ten items in the queue stored at *entry[2]* through *entry[11]*. The array size is 12. Where does the insert method place the new entry in the array? *entry[0]*

4. (2 points) If a queue user has a queue (*myQ*) that has 20 elements, why he should not clear all the queue contents using the function *create* as following?
as the space of the 20 elements is still reserved in the memory

creatQueue(&myQ);

5. (3 points) Consider an array based queue is declared as follows:

```
#define Max 20  
typedef char entryType;  
typedef struct  
    entryType entry[Max];  
    entryType *front;  
    entryType *rear;  
} queueType;  
int size;
```

Complete the following C function to initialize the queue to be empty.

*void creatQ(queueType *q){*

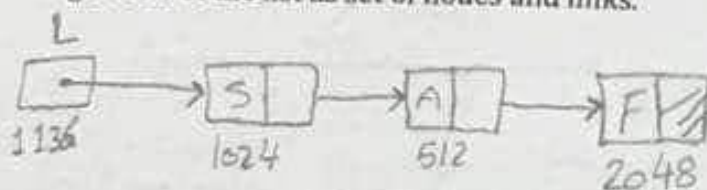
q->front = 2(q->entry[0]);

q->rear = 2(q->entry[Max-1]);

q->size = 0;

queueType myQ;

6. (3 points) Figure 1 is a snap shot for the memory that has a linked list of characters. If the head of the list is found in memory address 1136, draw the logic view of the list as set of nodes and links.



3

7. (6 points) Suppose you want to implement an ADT which contains two stacks in a single array, such that, the two stacks start from the two ends of the array and grow in opposite directions (towards one another). It is required to propose a suitable declaration part to implement that ADT and to define isFull function.

```

#define MAX 10
typedef int Entry_type;
typedef struct {
    Entry_type entry[MAX];
    int top1;
    int top2;
} New_type;

int isFull(New_type s) {
    return (s.top2 - s.top1 == 1);
}
  
```

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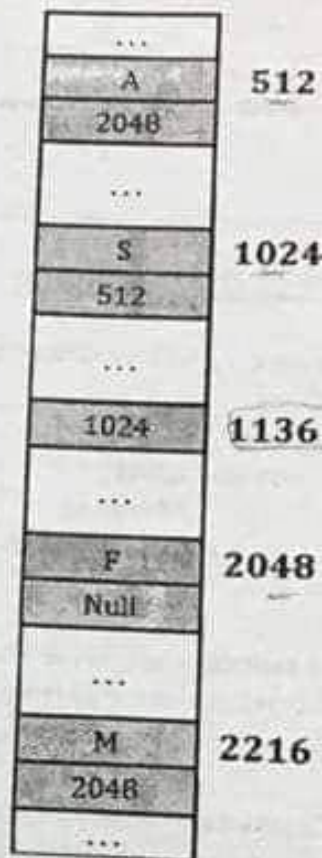


Figure 1

